
ME 605: Air Conditioning and Ventilation

Course Code	ME 605
Course Name	Air Conditioning and Ventilation
L-T-P-C	3-0-0-3
Prerequisites	Thermodynamics, heat transfer
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Fundamentals**

Importance of measurement and experimentation, calibration, uncertainty analysis, error propagation, Gaussian or Normal distribution, confidence level, regression analysis, correlation coefficient, Chi-Square test, zeroth-, first- and second-order systems. [Lectures]

- **Pressure Measurement**

Air properties, psychrometry, basic processes, and summer and winter air-conditioning. [Lectures]

- **Comfort**

Concept of human comfort and thermal response, comfort factors and environmental indices, Indoor Air Quality. [Lectures]

- **Equipment analysis and selection**

Filter, fan, air-washer, cooling tower and cooling and dehumidifying coil. [Lectures]

- **Load estimation**

Heat and mass transfer in wall, insulation, vapour barrier, and cooling and heating load calculations. [Lectures]

- **Air distribution**

Air flow in duct, duct sizing and space air diffusion. [Lectures]

- **Ventilation**

Methods, applications in industries, exhaust systems and design.

Text and Reference Books

1. A SHRAE Handbook
2. C.P. Arora, **Refrigeration and Airconditioning**, 3rd Edition, Tata McGraw-Hill 2009
3. R.C. Arora, **Refrigeration and Airconditioning**, 1st Edition, Prentice Hall India, 2010.
4. W. P. Jones, **Air conditioning engineering**, 5th Edition, Elsevier, 2001.